

Chapter 1: Introduction to Derivatives Markets (Continued)

3. Forward Contracts (Section 1.3)

a. Definition and Trading

- **Definition:** A **forward contract** is an agreement between two parties to buy or sell an asset at a **certain future time** for a **certain price** (the delivery price, K).
- **Contrast with Spot Contract:** A **spot contract** is an agreement to buy or sell an asset almost **immediately** (for immediate delivery).
- **Trading Market:** Forward contracts are traded in the **Over-the-Counter (OTC) market**, typically between two financial institutions or between a bank and a client.
- **Positions:**
 - **Long Position:** The party who **agrees to buy** the underlying asset on the specified future date at the price K .
 - **Short Position:** The party who **agrees to sell** the underlying asset on the specified future date at the price K .
- **Commitment:** Both parties enter into a **binding commitment** to fulfill the transaction.

b. Example: Foreign Exchange Forward Contract

- **Popularity:** Forward contracts on foreign exchange (FX) are very popular, with large banks employing both spot and forward FX traders.
- **Example (Hedging):** A US corporation knows it must pay £1 million in 6 months. To hedge the risk of the British pound (GBP) appreciating (costing more USD), the treasurer enters a **long forward contract** to buy £1 million in 6 months at a locked-in rate (e.g., \$1.5532/GBP).
 - **Corporation:** Takes a **long forward contract** (agrees to buy £1 million for \$1,553,200).
 - **Bank:** Takes a **short forward contract** (agrees to sell £1 million for \$1,553,200).

c. Payoffs from Forward Contracts

- The payoff is calculated based on the difference between the spot price at maturity (S_T) and the agreed-upon delivery price (K).
- **Long Position Payoff (Buyer):**

$$\text{Payoff} = S_T - K$$

- The holder is obligated to buy an asset worth S_T for a price K . The payoff is positive if $S_T > K$.
- **Short Position Payoff (Seller):**

$$\text{Payoff} = K - S_T$$

- The holder is obligated to sell an asset worth S_T for a price K . The payoff is positive if $K > S_T$.
- **Cost and Gain:** Since it costs **nothing** to enter into a standard forward contract, the payoff is also the trader's total gain or loss.

d. Relationship between Forward and Spot Prices

- **Arbitrage Principle:** Forward prices are linked to spot prices through interest rates, due to the opportunity for arbitrage (locking in a riskless profit).
- **Example (Non-Dividend Paying Stock):**
 - Spot Price (S_0): \$60.
 - 1-year Interest Rate (r): 5%.
 - The theoretical 1-year Forward Price (F_0) should be the spot price compounded by the interest rate: $F_0 = \$60 \times (1 + 0.05) = \63 .
 - **If $F_0 > \$63$ (e.g., \$67):** An arbitrageur can borrow \$60, buy the stock spot, and sell it forward for \$67. They repay \$63, netting a profit of \$4.
 - **If $F_0 < \$63$ (e.g., \$58):** An investor can sell their existing stock for \$60, invest the proceeds to earn \$3 (total \$63), and enter a forward contract to buy the stock back for \$58, netting a profit of \$5.

4. Futures Contracts (Section 1.4)

a. Definition and Key Differences from Forwards

- **Definition:** Like a forward contract, a **futures contract** is an agreement to buy or sell an asset at a certain future time for a certain price.
- **Trading Market:** Futures contracts are normally traded on a **Derivatives Exchange** (e.g., CME Group, formerly CBOT and CME).
- **Standardization:** To facilitate trading, the exchange specifies **standardized features** of the contract (quantity, quality, delivery date).
- **Guarantee:** The exchange provides a mechanism (the Clearing House) that guarantees the contract will be honored, as the two parties do not necessarily know each other.

b. Examples of Underlying Assets

- **Commodities:** Pork bellies, live cattle, sugar, wool, lumber, copper, aluminum, gold, tin, etc.
- **Financial Assets:** Stock indices, currencies, Treasury bonds, etc.
- **Futures Price:** The price quoted (e.g., December gold futures at \$1,380) is the price at which traders agree to buy or sell the asset for future delivery, determined by supply and demand.

c. Operational Details (See Chapter 2)

- Detailed operational issues like margin requirements, daily settlement procedures, delivery procedures, bid-offer spreads, and the role of the clearing house are covered extensively in Chapter 2.

5. Options (Section 1.5)

a. Definition and Key Terms

- **Trading Market:** Options are traded on both **Exchanges** (e.g., CBOE) and in the **OTC market**.
- **Option vs. Forward/Futures:** An option gives the holder the **right** to do something; they are **not obligated**. Forward and futures contracts create an **obligation**.
- **Cost:** There is a **cost (premium)** to acquiring an option, unlike forward/futures contracts.
- **Key Terms:**
 - **Exercise Price (or Strike Price, K):** The price specified in the contract at which the asset can be bought or sold.
 - **Expiration Date (or Maturity):** The date specified in the contract by which the option must be exercised.

b. Types of Options

Type	Right Granted
Call Option	Gives the holder the right to buy the underlying asset.
Put Option	Gives the holder the right to sell the underlying asset.

c. Exercise Styles

Style	Exercise Rule	Prevalance
American Option	Can be exercised at any time up to the expiration date.	Most options traded on exchanges are American.
European Option	Can be exercised only on the expiration date itself.	Easier to analyze; used as a basis for pricing American options.

d. Example and Payoffs

- **Exchange Contracts:** In the exchange-traded equity option market, one contract usually covers **100 shares**.
- **Call Option Example:** Buying a December call on Google with $K = \$880$ for a premium of \$56.30/share (total cost: \$5,630).
 - **Outcome if $S_T \leq \$880$:** Option expires worthless; loss equals the premium (\$5,630).
 - **Outcome if $S_T > \$880$ (e.g., $S_T = \$1,000$):** Option is exercised; gross profit is $(\$1,000 - \$880) \times 100 = \$12,000$. Net profit is $\$12,000 - \$5,630 = \$6,370$.
- **Put Option Example:** Selling (writing) a September put on Google with $K = \$840$ for a premium of \$31.00/share (immediate inflow: \$3,100).
 - **Outcome if $S_T \geq \$840$:** Option expires worthless; profit equals the premium (\$3,100).
 - **Outcome if $S_T < \$840$ (e.g., $S_T = \$800$):** Option is exercised against the seller; gross loss is $(\$840 - \$800) \times 100 = \$4,000$. Net loss is $\$4,000 - \$3,100 = \$900$.

e. Option Characteristics

- **Call Price vs. Strike Price:** The price of a call option **decreases** as the strike price (K) increases.
- **Put Price vs. Strike Price:** The price of a put option **increases** as the strike price (K) increases.
- **Maturity:** Both types of options generally become **more valuable** as their time to maturity increases.

f. Participants in Options Markets

1. **Buyers of Calls (Long Calls)**
2. **Sellers of Calls (Short Calls)** → Also known as **writing the option**.
3. **Buyers of Puts (Long Puts)**
4. **Sellers of Puts (Short Puts)** → Also known as **writing the option**.